

Протокол

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GitHub проєкту: https://github.com/bilyiandrii/soft_arch_hw_bilyi.git

Гілка: hz_distr_map

Скрипти знаходяться у папці **scripts**. Скрипти запускаються від лиця facade сервісу.

Крок 1

- Встановив Hazelcast через Docker.
- Використовую образ `hazelcast/hazelcast:5.3`.

```
docker pull hazelcast/hazelcast:5.3
```

```
5.3: Pulling from hazelcast/hazelcast
Digest: sha256:43e0214a3ab4e6e366f94fd8de7e980e90b0661a119ba17d7b7ac77472a8ea95
Status: Image is up to date for hazelcast/hazelcast:5.3
docker.io/hazelcast/hazelcast:5.3
```

- Налаштування зробив через `docker-compose.yml`.
- Також для взаємодії з кластером скачав бібліотеку `hazelcast-python-client`.

Крок 2

Приклад конфігурації для вузла та менеджмент центру:

```
hazelcast1:
  image: hazelcast/hazelcast:5.3
  ports:
    - "5701:5701"
  environment:
    - HZ_CLUSTERNAME=dev-cluster
```

Схожа механіка конфігурації використовується для кожного з вузлів. Вузли будуть об'єднуватись через спільну змінну середовища, `HZ_CLUSTERNAME=dev-cluster`.

```
management-center:
  image: hazelcast/management-center:5.3
  ports:
    - "8081:8080"
  environment:
    - MC_DEFAULT_CLUSTER=dev-cluster
    - MC_DEFAULT_CLUSTER_MEMBERS=hazelcast1,hazelcast2,hazelcast3
```

Менеджмент центр - GUI для перегляду стану кластера, який тут вказується явно через `MC_DEFAULT_CLUSTER` та для його вузлів - `MC_DEFAULT_CLUSTER_MEMBERS`.

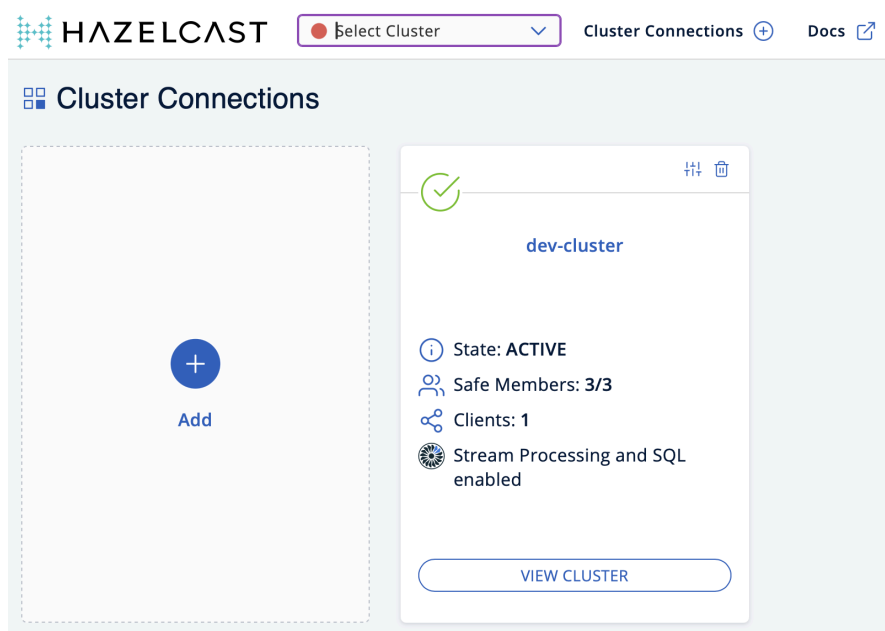
Щоб перевпентитись у розмірі кластера, можна вивести логи окремого вузла:

```
docker logs hws-hazelcast1-1
```

Бачимо що всі вони розпізнаються:

```
Members {size:3, ver:3} [  
  Member [172.18.0.5]:5701 - c0a4f489-1344-43ce-90c3-8f619c27966a  
  Member [172.18.0.2]:5701 - e64272a5-dfb3-4498-abb1-2aa6b340be0a this  
  Member [172.18.0.4]:5701 - f22dfaf7-ee4a-4b75-9660-2c8070eae483  
]
```

А якщо перейти на localhost:8081 то можна побачити наш кластер та його кількість учасників:



Крок 3

Імплементація скрипта знаходиться в step3_demo.py.

Логування будуть діставатися з третьої ноди:

```
docker logs -f hws-hazelcast3-1
```

Запускаю програму через фасад-сервіс, тобто всередині мережі docker:

```
docker compose exec facade uv run python step3_demo.py
```

Як бачимо запис відбувся успішно:

```
Connected to Hazelcast cluster.  
Writing 1000 values...  
Finished writing data. Check the Management Center!
```

Логи з докера як підтвердження:

```
2026-02-25 14:02:06,942 [ INFO] [hz.happy_zhukovsky.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=5, /172.18.0.7:5701->/172.18.0.8:48566, qualifier=null, endpoint=[172.18.0.8]:48566, remoteUid=b3067031-9bdc-4409-abd8-49ada071833b, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: b3067031-9bdc-4409-abd8-49ada071833b, client name: hz.client_0, client version: 5.6.0
2026-02-25 14:02:08,958 [ INFO] [hz.happy_zhukovsky.IO.thread-in-1] [c.h.i.s.t.TcpServerConnection]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connection[id=5, /172.18.0.7:5701->hws-facade-1.hws_default/172.18.0.8:48566, qualifier=null, endpoint=[172.18.0.8]:48566, remoteUid=b3067031-9bdc-4409-abd8-49ada071833b, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
2026-02-25 14:02:08,963 [ INFO] [hz.happy_zhukovsky.event-3] [c.h.c.i.ClientEndpointManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Destroying ClientEndpoint{connection=Connection[id=5, /172.18.0.7:5701->hws-facade-1.hws_default/172.18.0.8:48566, qualifier=null, endpoint=[172.18.0.8]:48566, remoteUid=b3067031-9bdc-4409-abd8-49ada071833b, alive=false, connectionType=PYH, planeIndex=-1], clientUid=b3067031-9bdc-4409-abd8-49ada071833b, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772028126942, latest clientAttributes=null, labels={}}
```

Розподіл ключів по нодах:

Map Statistics (In-Memory Format: BINARY) RESET TIME 1 minute ago → now Default View 🗑️ 📄 +

Member ^	Entries ^	Gets ^	Puts ^	Removals ^	Sets ^	Entry Memory ^	Ev. ^
172.18.0.5:5701	332	0	332	0	0	44.02 kB	
172.18.0.6:5701	321	0	321	0	0	42.56 kB	
172.18.0.7:5701	347	0	347	0	0	46.01 kB	
TOTAL	1,000	0	1,000	0	0	132.60 kB	

1 - 3 of 3 Rows 10

Як видно, записи були розподілені приблизно рівномірно між кожним вузлом.

Тепер, спробуємо відключити одну ноду:

```
docker stop hws-hazelcast1-1
```

```
2026-02-25 14:08:11,293 [ INFO] [hz.happy_zhukovsky.IO.thread-in-1] [c.h.i.s.t.TcpServerConnection]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connection[id=2, /172.18.0.7:5701->/172.18.0.5:51341, qualifier=null, endpoint=[172.18.0.5]:5701, remoteUid=fe9edb08-4a3f-4b4a-88b2-c466453d2eb4, alive=false, connectionType=MEMBER, planeIndex=0] closed. Reason: Connection closed by the other side
2026-02-25 14:08:11,294 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.5:5701, timeout: 10000, bind-any: true
2026-02-25 14:08:11,295 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.5:5701. Reason: IOException[Connection refused to address /172.18.0.5:5701]
2026-02-25 14:08:11,397 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.5:5701, timeout: 10000, bind-any: true
2026-02-25 14:08:11,397 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.5:5701. Reason: IOException[Connection refused to address /172.18.0.5:5701]
2026-02-25 14:08:11,498 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.5:5701, timeout: 10000, bind-any: true
2026-02-25 14:08:21,506 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.5:5701. Reason: IOException[null to address /172.18.0.5:5701]
2026-02-25 14:08:21,608 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.5:5701, timeout: 10000, bind-any: true
2026-02-25 14:08:31,617 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.5:5701. Reason: IOException[null to address /172.18.0.5:5701]
2026-02-25 14:08:31,618 [ WARN] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Removing connection to endpoint [172.18.0.5]:5701 Cause => java.io.IOException {null to address /172.18.0.5:5701}, Error-Count: 5
2026-02-25 14:08:31,620 [ WARN] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Member [172.18.0.5]:5701 - fe9edb08-4a3f-4b4a-88b2-c466453d2eb4 is suspected to be dead for reason: No connection
2026-02-25 14:08:31,626 [ INFO] [hz.happy_zhukovsky.priority-generic-operation.thread-0] [c.h.i.p.i.PartitionStateManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Storing snapshot of partition assignments while removing UUID fe9edb08-4a3f-4b4a-88b2-c466453d2eb4
2026-02-25 14:08:31,629 [ INFO] [hz.happy_zhukovsky.priority-generic-operation.thread-0] [c.h.i.c.ClusterService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8]
Members {size:2, ver:4} [
  Member [172.18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770
  Member [172.18.0.7]:5701 - 7334d6c9-c88d-4cec-b93f-f9d089334b34 this
]
2026-02-25 14:08:31,636 [ INFO] [hz.happy_zhukovsky.cached.thread-2] [c.h.t.TransactionManagerService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Committing/rolling-back live transactions of [172.18.0.5]:5701, UUID: fe9edb08-4a3f-4b4a-88b2-c466453d2eb4
```

Тепер розмір кластеру з двох.

Як можна помітити, дані з першого вузла були розподілені між 2-им та 3-ім:

Map Statistics (In-Memory Format: BINARY)

RESET TIME

1 minute ago

→

now

Default View



Member	Entries	Gets	Puts	Removals	Sets	Entry Memory	Ev
172.18.0.6:5701	476	0	321	0	0	63.11 kB	
172.18.0.7:5701	524	0	347	0	0	69.49 kB	
TOTAL	1,000	0	668	0	0	132.60 kB	

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А якщо відключити ще одну?:

docker stop hws-hazelcast2-1

```
509d88-1efc-4a5b-a5af-aa022c0f1770, alive=false, connectionType=MEMBER, planeIndex=0] closed. Reason: Connection closed by the other side
2026-02-25 14:10:52,668 [ INFO] [hz.happy_zhukovsky.cached.thread-2] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cl
uster] [5.3.8] Connecting to /172.18.0.6:5701, timeout: 10000, bind-any: true
2026-02-25 14:10:52,669 [ INFO] [hz.happy_zhukovsky.cached.thread-2] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cl
uster] [5.3.8] Could not connect to: /172.18.0.6:5701. Reason: IOException[Connection refused to address /172.18.0.6:5701]
2026-02-25 14:10:52,771 [ INFO] [hz.happy_zhukovsky.cached.thread-12] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-c
luster] [5.3.8] Connecting to /172.18.0.6:5701, timeout: 10000, bind-any: true
2026-02-25 14:10:52,772 [ INFO] [hz.happy_zhukovsky.cached.thread-12] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-c
luster] [5.3.8] Could not connect to: /172.18.0.6:5701. Reason: IOException[Connection refused to address /172.18.0.6:5701]
2026-02-25 14:10:52,873 [ INFO] [hz.happy_zhukovsky.cached.thread-12] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-c
luster] [5.3.8] Connecting to /172.18.0.6:5701, timeout: 10000, bind-any: true
2026-02-25 14:11:02,890 [ INFO] [hz.happy_zhukovsky.cached.thread-12] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-c
luster] [5.3.8] Could not connect to: /172.18.0.6:5701. Reason: IOException[null to address /172.18.0.6:5701]
2026-02-25 14:11:41,694 [ WARN] [hz.happy_zhukovsky.cached.thread-12] [c.h.i.c.i.ClusterHeartbeatManager]: [172.18.0.7]:5701 [
dev-cluster] [5.3.8] This node does not have a connection to Member [172.18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770
2026-02-25 14:11:46,694 [ WARN] [hz.happy_zhukovsky.cached.thread-8] [c.h.i.c.i.ClusterHeartbeatManager]: [172.18.0.7]:5701 [d
ev-cluster] [5.3.8] This node does not have a connection to Member [172.18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770
2026-02-25 14:11:49,413 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cl
uster] [5.3.8] Connecting to /172.18.0.6:5701, timeout: 10000, bind-any: true
2026-02-25 14:11:51,694 [ WARN] [hz.happy_zhukovsky.cached.thread-4] [c.h.i.c.i.ClusterHeartbeatManager]: [172.18.0.7]:5701 [d
ev-cluster] [5.3.8] This node does not have a connection to Member [172.18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770
2026-02-25 14:11:51,699 [ WARN] [hz.happy_zhukovsky.cached.thread-4] [c.h.i.c.i.ClusterHeartbeatManager]: [172.18.0.7]:5701 [d
ev-cluster] [5.3.8] Suspecting Member [172.18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770 because it has not sent any hea
rtbeats since 2026-02-25 14:10:51.681. Now: 2026-02-25 14:11:51.694, heartbeat timeout: 60000 ms, suspicion level: 1.00
2026-02-25 14:11:51,700 [ WARN] [hz.happy_zhukovsky.cached.thread-4] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-clu
ster] [5.3.8] Member [172.18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770 is suspected to be dead for reason: Suspecting M
ember [172.18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770 because it has not sent any heartbeats since 2026-02-25 14:10:5
1.681. Now: 2026-02-25 14:11:51.694, heartbeat timeout: 60000 ms, suspicion level: 1.00
2026-02-25 14:11:51,700 [ INFO] [hz.happy_zhukovsky.cached.thread-4] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-clu
ster] [5.3.8] Starting mastership claim process...
2026-02-25 14:11:51,700 [ INFO] [hz.happy_zhukovsky.cached.thread-4] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-clu
ster] [5.3.8] local MembersView(version=4, members=[MemberInfo{address=[172.18.0.6]:5701, uuid=8c509d88-1efc-4a5b-a5af-aa022c0
f1770, cpMemberUUID=null, liteMember=false, memberListJoinVersion=1}, MemberInfo{address=[172.18.0.7]:5701, uuid=7334dc6c9-c88d
-4cec-b93f-f9d089334b34, cpMemberUUID=null, liteMember=false, memberListJoinVersion=2}}] with suspected members: [Member [172.
18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770] and initial addresses to ask: []
2026-02-25 14:11:51,702 [ INFO] [hz.happy_zhukovsky.cached.thread-2] [c.h.i.p.i.PartitionStateManager]: [172.18.0.7]:5701 [dev
-cluster] [5.3.8] Storing snapshot of partition assignments while removing UUID 8c509d88-1efc-4a5b-a5af-aa022c0f1770
2026-02-25 14:11:51,702 [ INFO] [hz.happy_zhukovsky.cached.thread-2] [c.h.i.c.ClusterService]: [172.18.0.7]:5701 [dev-cluster]
[5.3.8]
Members {size:1, ver:5} [
  Member [172.18.0.7]:5701 - 7334dc6c9-c88d-4cec-b93f-f9d089334b34 this
]
2026-02-25 14:11:51,704 [ INFO] [hz.happy_zhukovsky.cached.thread-2] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-clu
ster] [5.3.8] Mastership is claimed with: MembersView{version=5, members=[MemberInfo{address=[172.18.0.7]:5701, uuid=7334dc6c9-
c88d-4cec-b93f-f9d089334b34, cpMemberUUID=null, liteMember=false, memberListJoinVersion=2}}]
2026-02-25 14:11:51,704 [ INFO] [hz.happy_zhukovsky.cached.thread-7] [c.h.t.TransactionManagerService]: [172.18.0.7]:5701 [dev
-cluster] [5.3.8] Committing/rolling-back live transactions of [172.18.0.6]:5701, UUID: 8c509d88-1efc-4a5b-a5af-aa022c0f1770
2026-02-25 14:11:51,709 [ INFO] [hz.happy_zhukovsky.migration] [c.h.i.p.InternalPartitionService]: [172.18.0.7]:5701 [dev-clus
ter] [5.3.8] Fetching partition tables from cluster to determine the most recent one... Local stamp: 6082267772165243149
2026-02-25 14:11:51,709 [ INFO] [hz.happy_zhukovsky.migration] [c.h.i.p.InternalPartitionService]: [172.18.0.7]:5701 [dev-clus
ter] [5.3.8] Most recent partition table is determined.
2026-02-25 14:11:51,709 [ INFO] [hz.happy_zhukovsky.migration] [c.h.i.p.InternalPartitionService]: [172.18.0.7]:5701 [dev-clus
ter] [5.3.8] Applying the most recent of partition state...
2026-02-25 14:11:51,710 [ WARN] [hz.happy_zhukovsky.migration] [c.h.i.p.InternalPartitionService]: [172.18.0.7]:5701 [dev-clus
ter] [5.3.8] Following unknown addresses are found in partition table sent from master[[172.18.0.7]:5701]. (Probably they have
recently joined or left the cluster.) {
  [172.18.0.6]:5701 - 8c509d88-1efc-4a5b-a5af-aa022c0f1770
}
2026-02-25 14:11:51,776 [ INFO] [hz.happy_zhukovsky.migration] [c.h.i.p.i.MigrationManager]: [172.18.0.7]:5701 [dev-cluster] [
5.3.8] Partition balance is ok, no need to repartition.
2026-02-25 14:11:52,510 [ INFO] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cl
uster] [5.3.8] Could not connect to: /172.18.0.6:5701. Reason: IOException[Host is unreachable to address /172.18.0.6:5701]
2026-02-25 14:11:52,511 [ WARN] [hz.happy_zhukovsky.cached.thread-1] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [172.18.0.7]
:5701 [dev-cluster] [5.3.8] Removing connection to endpoint [172.18.0.6]:5701 Cause => java.io.IOException {Host is unreachabl
e to address /172.18.0.6:5701}, Error-Count: 5
```

Як і передбачалося, дані були перекинуті на останню активну ноду.

Map Statistics (In-Memory Format: BINARY)

RESET TIME

1 minute ago

→

now

Default View



Member	Entries	Gets	Puts	Removals	Sets	Entry Memory	Ev
172.18.0.7:5701	1,000	0	347	0	0	132.60 kB	
TOTAL	1,000	0	347	0	0	132.60 kB	

1 - 1 of 1 Rows10

А якщо одночасно дві?

```
docker kill -s SIGKILL hws-hazelcast1-1 hws-hazelcast2-1
```

Логи

```
0e963e43-8c0b-4f21-bccd-d3df53aa7b80, alive=false, connectionType=MEMBER, planeIndex=0] closed. Reason: Connection closed by the other side
2026-02-25 14:16:31,581 [ INFO] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.4:5701, timeout: 10000, bind-any: true
2026-02-25 14:16:31,581 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.2:5701, timeout: 10000, bind-any: true
2026-02-25 14:16:31,582 [ INFO] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.4:5701. Reason: IOException[Connection refused to address /172.18.0.4:5701]
2026-02-25 14:16:31,582 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.2:5701. Reason: IOException[Connection refused to address /172.18.0.2:5701]
2026-02-25 14:16:31,683 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.2:5701, timeout: 10000, bind-any: true
2026-02-25 14:16:31,683 [ INFO] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.4:5701, timeout: 10000, bind-any: true
2026-02-25 14:16:31,684 [ INFO] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.4:5701. Reason: IOException[Connection refused to address /172.18.0.4:5701]
2026-02-25 14:16:31,785 [ INFO] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.4:5701, timeout: 10000, bind-any: true
2026-02-25 14:16:41,699 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.2:5701. Reason: IOException[null to address /172.18.0.2:5701]
2026-02-25 14:16:41,801 [ INFO] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.4:5701. Reason: IOException[null to address /172.18.0.4:5701]
2026-02-25 14:16:41,801 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.2:5701, timeout: 10000, bind-any: true
2026-02-25 14:16:41,904 [ INFO] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.4:5701, timeout: 10000, bind-any: true
2026-02-25 14:16:51,810 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.2:5701. Reason: IOException[null to address /172.18.0.2:5701]
2026-02-25 14:16:51,912 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Connecting to /172.18.0.2:5701, timeout: 10000, bind-any: true
2026-02-25 14:16:51,912 [ INFO] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.4:5701. Reason: IOException[null to address /172.18.0.4:5701]
2026-02-25 14:16:51,913 [ WARN] [hz.youthful_torvalds.cached.thread-3] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Removing connection to endpoint [172.18.0.4]:5701 Cause => java.io.IOException [null to address /172.18.0.4:5701], Error-Count: 5
2026-02-25 14:16:51,914 [ WARN] [hz.youthful_torvalds.cached.thread-3] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Member [172.18.0.4]:5701 - 0e963e43-8c0b-4f21-bccd-d3df53aa7b80 is suspected to be dead for reason: No connection
2026-02-25 14:17:01,925 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Could not connect to: /172.18.0.2:5701. Reason: IOException[null to address /172.18.0.2:5701]
2026-02-25 14:17:01,926 [ WARN] [hz.youthful_torvalds.cached.thread-5] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Removing connection to endpoint [172.18.0.2]:5701 Cause => java.io.IOException [null to address /172.18.0.2:5701], Error-Count: 5
2026-02-25 14:17:01,926 [ WARN] [hz.youthful_torvalds.cached.thread-5] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Member [172.18.0.2]:5701 - ba0670da-7135-448b-b5f5-62bf5038bbc8 is suspected to be dead for reason: No connection
2026-02-25 14:17:01,926 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Starting mastership claim process...
2026-02-25 14:17:01,927 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Local MembersView{version=3, members=[MemberInfo{address=[172.18.0.2]:5701, uuid=ba0670da-7135-448b-b5f5-62bf5038bbc8, cpMemberUUID=null, liteMember=false, memberListJoinVersion=1}, MemberInfo{address=[172.18.0.4]:5701, uuid=0e963e43-8c0b-4f21-bccd-d3df53aa7b80, cpMemberUUID=null, liteMember=false, memberListJoinVersion=2}, MemberInfo{address=[172.18.0.7]:5701, uuid=f7116010-d52a-4018-9c33-27af61c05bea, cpMemberUUID=null, liteMember=false, memberListJoinVersion=3}]} with suspected members: [Member [172.18.0.4]:5701 - 0e963e43-8c0b-4f21-bccd-d3df53aa7b80, Member [172.18.0.2]:5701 - ba0670da-7135-448b-b5f5-62bf5038bbc8] and initial addresses to ask: []
2026-02-25 14:17:01,930 [ INFO] [hz.youthful_torvalds.cached.thread-1] [c.h.i.p.i.PartitionStateManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Storing snapshot of partition assignments while removing UUID ba0670da-7135-448b-b5f5-62bf5038bbc8
2026-02-25 14:17:01,937 [ INFO] [hz.youthful_torvalds.cached.thread-1] [c.h.i.p.i.PartitionStateManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Storing snapshot of partition assignments while removing UUID 0e963e43-8c0b-4f21-bccd-d3df53aa7b80
2026-02-25 14:17:01,941 [ INFO] [hz.youthful_torvalds.cached.thread-1] [c.h.i.c.ClusterService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8]

Members {size:1, ver:4} [
  Member [172.18.0.7]:5701 - f7116010-d52a-4018-9c33-27af61c05bea this
]
```

```
2026-02-25 14:17:01,942 [ INFO] [hz.youthful_torvalds.cached.thread-1] [c.h.i.c.i.MembershipManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Mastership is claimed with: MembersView{version=4, members=[MemberInfo{address=[172.18.0.7]:5701, uuid=f7116010-d52a-4018-9c33-27af61c05bea, cpMemberUUID=null, liteMember=false, memberListJoinVersion=3}]}
2026-02-25 14:17:01,942 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.t.TransactionManagerService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Committing/rolling-back live transactions of [172.18.0.2]:5701, UUID: ba0670da-7135-448b-b5f5-62bf5038bbc8
2026-02-25 14:17:01,947 [ INFO] [hz.youthful_torvalds.cached.thread-5] [c.h.t.TransactionManagerService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Committing/rolling-back live transactions of [172.18.0.4]:5701, UUID: 0e963e43-8c0b-4f21-bccd-d3df53aa7b80
2026-02-25 14:17:02,188 [ INFO] [hz.youthful_torvalds.migration] [c.h.i.p.InternalPartitionService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Fetching partition tables from cluster to determine the most recent one... Local stamp: 808208589080019451
2026-02-25 14:17:02,188 [ INFO] [hz.youthful_torvalds.migration] [c.h.i.p.InternalPartitionService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Most recent partition table is determined.
2026-02-25 14:17:02,188 [ INFO] [hz.youthful_torvalds.migration] [c.h.i.p.InternalPartitionService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Applying the most recent of partition state...
2026-02-25 14:17:02,189 [ WARN] [hz.youthful_torvalds.migration] [c.h.i.p.InternalPartitionService]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Following unknown addresses are found in partition table sent from master[172.18.0.7]:5701. (Probably they have recently joined or left the cluster.) {
  [172.18.0.4]:5701 - 0e963e43-8c0b-4f21-bccd-d3df53aa7b80
  [172.18.0.2]:5701 - ba0670da-7135-448b-b5f5-62bf5038bbc8
}
2026-02-25 14:17:02,257 [ INFO] [hz.youthful_torvalds.migration] [c.h.i.p.i.MigrationManager]: [172.18.0.7]:5701 [dev-cluster] [5.3.8] Partition balance is ok, no need to repartition.
```

Видно, що розмір кластеру зменшився до одного та повідомлення Following unknown addresses are found... [Probably they have recently joined or left the cluster.]

Зі зображення видно, що третина записів була втрачена:

Map Statistics (In-Memory Format: BINARY) RESET TIME 1 minute ago → now Default View 🗑️ 📄 +

Member ^	Entries ^	Gets ^	Puts ^	Removals ^	Sets ^	Entry Memory ^	Ev ^
172.18.0.7:5701	643	0	316	0	0	85.28 kB	
TOTAL	643	0	316	0	0	85.28 kB	

1 - 1 of 1 Rows 10 ▼

За замовчуванням Hazelcast створює одну резервну копію для кожного запису, тобто `backup-count=1` [\[1\]](#).

Map Config

Backup Count:	Async Backup Count:	In-Memory Storage Format:
1	0	BINARY
Max Size:	Max Size Policy:	Max Idle (seconds):
2147483647	PER_NODE	0
TTL (seconds):	Eviction Policy:	☐ Read Backup Data
0	NONE	
Merge Policy:		
com.hazelcast.spi.merge.PutIfAbsentMergePolicy		

[Click here](#) for more information about configuration options

[Cancel](#)

Звідси випливає, що при відключенні одного вузла чи послідовно двох - втрати даних не буде. В цих ситуаціях, вцілілі ноди матимуть час на відновлення та створення нових бекапів, на відміну від одночасного падіння серверів. Тут, бекап однієї з вбитих нод міг лежати на іншій, яка теж вийшла з ладу. Через те, що вони впали миттєво і одночасно, то кластер не встиг зреагувати та відновити копію. А оскільки Hazelcast є in-memory платформою, то після падіння сервера дані на ньому повністю втрачаються.

Щоб уникнути втрати даних потрібно збільшити кількість резервних копій (`backup-count=2`). Таким чином кожен запис матиме 1 оригінал та 2 бекапи, тобто копія всіх даних буде зберігатися на кожному з трьох вузлів. Конфігурація робиться через `hazelcast.xml` або `hazelcast.yaml`.

```
hazelcast:
  cluster-name: dev-cluster
  network:
    join:
      tcp-ip:
        enabled: true
        member-list:
          - hazelcast1
          - hazelcast2
          - hazelcast3
  map:
```

```
step3_demo_map:
  backup-count: 2
  async-backup-count: 0
```

Тут я використовую комунікацію для вузлів через tcp/ip замість multicast, так би мовити хардкоджу адреси кожного з нод (member-list). Нижче вже йде налаштування окремих структур, в нашому випадку для map.

docker-compose.yaml:

```
volumes:
  - ./hazelcast.yaml:/opt/hazelcast/config/hazelcast.yaml
environment:
  - HZ_CLUSTERNAME=dev-cluster
  - JAVA_OPTS=-Dhazelcast.config=/opt/hazelcast/config/hazelcast.yaml
```

Щоб скопіювати цей файл до контейнер, треба явно вказати його розташування локально на машині. Потім за допомогою змінної **JAVA_OPTS** чітко вказати Hazelcast використовувати цей файл замість типового.

Бачимо що кількість бекапів змінилась:

Map Config

Backup Count:	Async Backup Count:	In-Memory Storage Format:
2	0	BINARY
Max Size:	Max Size Policy:	Max Idle (seconds):
2147483647	PER_NODE	0
TTL (seconds):	Eviction Policy:	☐ Read Backup Data
0	NONE	
Merge Policy:		
com.hazelcast.spi.merge.PutIfAbsentMergePolicy		

[Click here for more information about configuration options](#) [Cancel](#)

Ось логи з докера після одночасного анулювання серверів:

```
andrii@myprookie hws % docker logs -f hws-hazelcast3-1
er] [5.3.8] Connecting to hazelcast1/172.18.0.7:5701, timeout: 10000, bind-any: true
2026-02-25 14:32:54,832 [ INFO] [hz.hungry_ride.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Connecting to hazelcast2/172.18.0.5:5701, timeout: 10000, bind-any: true
2026-02-25 14:32:54,833 [ INFO] [hz.hungry_ride.cached.thread-2] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Connecting to hazelcast1/172.18.0.7:5701, timeout: 10000, bind-any: true
2026-02-25 14:32:54,833 [ INFO] [hz.hungry_ride.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Could not connect to: hazelcast1/172.18.0.7:5701. Reason: IOException[Connection refused to address hazelcast1/17
2.18.0.7:5701]
2026-02-25 14:32:54,833 [ INFO] [hz.hungry_ride.cached.thread-2] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Could not connect to: hazelcast1/172.18.0.7:5701. Reason: IOException[Connection refused to address hazelcast1/17
2.18.0.7:5701]
2026-02-25 14:32:54,833 [ INFO] [hz.hungry_ride.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Could not connect to: hazelcast2/172.18.0.5:5701. Reason: IOException[Connection refused to address hazelcast2/17
2.18.0.5:5701]
2026-02-25 14:32:54,935 [ INFO] [hz.hungry_ride.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Connecting to hazelcast2/172.18.0.5:5701, timeout: 10000, bind-any: true
2026-02-25 14:32:54,935 [ INFO] [hz.hungry_ride.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Connecting to hazelcast1/172.18.0.7:5701, timeout: 10000, bind-any: true
2026-02-25 14:32:54,935 [ INFO] [hz.hungry_ride.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Could not connect to: hazelcast1/172.18.0.7:5701. Reason: IOException[Connection refused to address hazelcast1/17
2.18.0.7:5701]
2026-02-25 14:32:55,038 [ INFO] [hz.hungry_ride.cached.thread-4] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Connecting to hazelcast1/172.18.0.7:5701, timeout: 10000, bind-any: true
2026-02-25 14:33:04,947 [ INFO] [hz.hungry_ride.cached.thread-1] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Could not connect to: hazelcast2/172.18.0.5:5701. Reason: IOException[null to address hazelcast2/172.18.0.5:5701]
2026-02-25 14:33:05,044 [ INFO] [hz.hungry_ride.cached.thread-4] [c.h.i.s.t.TcpServerConnector]: [hazelcast3]:5701 [dev-clust
er] [5.3.8] Could not connect to: hazelcast1/172.18.0.7:5701. Reason: IOException[null to address hazelcast1/172.18.0.7:5701]
2026-02-25 14:33:25,232 [ WARN] [hz.hungry_ride.cached.thread-7] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [hazelcast3]:57
01 [dev-cluster] [5.3.8] Removing connection to endpoint [hazelcast2]:5701 Cause => java.net.UnknownHostException {hazelcast2
}, Error-Count: 5
2026-02-25 14:33:25,232 [ WARN] [hz.hungry_ride.cached.thread-3] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [hazelcast3]:57
01 [dev-cluster] [5.3.8] Removing connection to endpoint [hazelcast1]:5701 Cause => java.net.UnknownHostException {hazelcast1
}, Error-Count: 5
2026-02-25 14:33:25,233 [ WARN] [hz.hungry_ride.cached.thread-3] [c.h.i.c.i.MembershipManager]: [hazelcast3]:5701 [dev-cluste
r] [5.3.8] Member [hazelcast1]:5701 - dc6b4438-8587-4246-9159-fa8011a8411c is suspected to be dead for reason: No connection
2026-02-25 14:33:25,233 [ INFO] [hz.hungry_ride.cached.thread-3] [c.h.i.c.i.MembershipManager]: [hazelcast3]:5701 [dev-cluste
r] [5.3.8] Starting mastership claim process...

1 [dev-cluster] [5.3.8] Removing connection to endpoint [hazelcast1]:5701 Cause => java.net.UnknownHostException {hazelcast1
}, Error-Count: 52
2026-02-25 14:33:30,242 [ INFO] [hz.hungry_ride.cached.thread-8] [c.h.i.p.i.PartitionStateManager]: [hazelcast3]:5701 [dev-cl
ster] [5.3.8] Storing snapshot of partition assignments while removing UUID dc6b4438-8587-4246-9159-fa8011a8411c
2026-02-25 14:33:30,245 [ INFO] [hz.hungry_ride.cached.thread-8] [c.h.i.p.i.PartitionStateManager]: [hazelcast3]:5701 [dev-cl
ster] [5.3.8] Storing snapshot of partition assignments while removing UUID aa1343f6-478b-4848-8ab9-31128115a3d5
2026-02-25 14:33:30,245 [ INFO] [hz.hungry_ride.cached.thread-8] [c.h.i.c.ClusterService]: [hazelcast3]:5701 [dev-cluster] [5
3.8]

Members {size:1, ver:4} [
  Member [hazelcast3]:5701 - 6ac0c0b0-27f9-4436-8e01-cddfcf6916ec this

2026-02-25 14:33:30,246 [ INFO] [hz.hungry_ride.cached.thread-8] [c.h.i.c.i.MembershipManager]: [hazelcast3]:5701 [dev-cluste
r] [5.3.8] Mastership is claimed with: MembersView{version=4, members=[MemberInfo{address=[hazelcast3]:5701, uuid=6ac0c0b0-27
f9-4436-8e01-cddfcf6916ec, cpMemberUUID=null, liteMember=false, memberListJoinVersion=2}}]
2026-02-25 14:33:30,248 [ INFO] [hz.hungry_ride.cached.thread-7] [c.h.t.TransactionManagerService]: [hazelcast3]:5701 [dev-cl
ster] [5.3.8] Committing/rolling-back live transactions of [hazelcast1]:5701, UUID: dc6b4438-8587-4246-9159-fa8011a8411c
2026-02-25 14:33:30,250 [ INFO] [hz.hungry_ride.cached.thread-7] [c.h.t.TransactionManagerService]: [hazelcast3]:5701 [dev-cl
ster] [5.3.8] Committing/rolling-back live transactions of [hazelcast2]:5701, UUID: aa1343f6-478b-4848-8ab9-31128115a3d5
2026-02-25 14:33:30,347 [ INFO] [hz.hungry_ride.migration] [c.h.i.p.InternalPartitionService]: [hazelcast3]:5701 [dev-cluster
] [5.3.8] Fetching partition tables from cluster to determine the most recent one... Local stamp: 8082085890800019451
2026-02-25 14:33:30,347 [ INFO] [hz.hungry_ride.migration] [c.h.i.p.InternalPartitionService]: [hazelcast3]:5701 [dev-cluster
] [5.3.8] Most recent partition table is determined.
2026-02-25 14:33:30,347 [ INFO] [hz.hungry_ride.migration] [c.h.i.p.InternalPartitionService]: [hazelcast3]:5701 [dev-cluster
] [5.3.8] Applying the most recent of partition state...
2026-02-25 14:33:30,348 [ WARN] [hz.hungry_ride.migration] [c.h.i.p.InternalPartitionService]: [hazelcast3]:5701 [dev-cluster
] [5.3.8] Following unknown addresses are found in partition table sent from master[hazelcast3]:5701. (Probably they have r
ecently joined or left the cluster.) {
  [hazelcast1]:5701 - dc6b4438-8587-4246-9159-fa8011a8411c
  [hazelcast2]:5701 - aa1343f6-478b-4848-8ab9-31128115a3d5

2026-02-25 14:33:30,437 [ INFO] [hz.hungry_ride.migration] [c.h.i.p.i.MigrationManager]: [hazelcast3]:5701 [dev-cluster] [5.3
8] Partition balance is ok, no need to repartition.
2026-02-25 14:36:37,832 [ WARN] [hz.hungry_ride.cached.thread-2] [c.h.i.c.i.TcpIpJoiner]: [hazelcast3]:5701 [dev-cluster] [5.
3.8] Cannot resolve hostname 'hazelcast2'. Please make sure host is valid and reachable.
2026-02-25 14:36:37,838 [ WARN] [hz.hungry_ride.cached.thread-2] [c.h.i.c.i.TcpIpJoiner]: [hazelcast3]:5701 [dev-cluster] [5.
3.8] Cannot resolve hostname 'hazelcast1'. Please make sure host is valid and reachable.
2026-02-25 14:38:37,827 [ WARN] [hz.hungry_ride.cached.thread-9] [c.h.i.c.i.TcpIpJoiner]: [hazelcast3]:5701 [dev-cluster] [5.
3.8] Cannot resolve hostname 'hazelcast2'. Please make sure host is valid and reachable.
2026-02-25 14:38:37,831 [ WARN] [hz.hungry_ride.cached.thread-9] [c.h.i.c.i.TcpIpJoiner]: [hazelcast3]:5701 [dev-cluster] [5.
3.8] Cannot resolve hostname 'hazelcast1'. Please make sure host is valid and reachable.
```

Спостерігаємо: Comittin/rolling-back live transactions of [hazelcast1(2)]. Отже дані спокійно відновлюються.

А ось як виглядає таблиця:

Map Statistics (In-Memory Format: BINARY) RESET TIME 1 minute ago → now Default View 🗑️ 📝 +

Member ^	^ Entries	^ Gets	^ Puts	^ Removals	^ Sets	^ Entry Memory	^ Ev
hazelcast1:5701	N/A	N/A	N/A	N/A	N/A	N/A	
hazelcast2:5701	N/A	N/A	N/A	N/A	N/A	N/A	
hazelcast3:5701	1,000	0	334	0	0	132.60 kB	
TOTAL	1,000	0	334	0	0	132.60 kB	

1 - 3 of 3 Rows 10 ▾

Тут ноди позначені як N/A, оскільки третя нода на них чекає (в надії що скоро повернуться).

Крок 4, 5, 6

- Використовую потоки для симуляцій одночасних запитів клієнтів
- Глобальні змінні: ITERATIONS = 10K (к-ть запитів клієнта).
- Скрипт виконується по одному разу для кожної зі ситуацій.
- Програма повертає кінцеве значення за ключем 'key' та час роботи.

Без блокувань:

```
def run_no_locks(client_id):
    for k in range(ITERATIONS):
        value = hz_map.get("key").result() or 0
        value += 1
        hz_map.put("key", value).result()
```

Песимістичне блокування:

```
def run_pessimistic(client_id):
    for k in range(ITERATIONS):
        hz_map.lock("key").result()
        try:
            value = hz_map.get("key").result() or 0
            value += 1
            hz_map.put("key", value).result()
        finally:
            hz_map.unlock("key").result()
```

Оптимістичне блокування:

```
def run_optimistic(client_id):
    for k in range(ITERATIONS):
        while True:
            old_value = hz_map.get("key").result() or 0
            new_value = old_value + 1
            if hz_map.replace_if_same("key", old_value, new_value).result():
                break
```

Команда

```
docker compose exec facade uv run python step456_locks.py
```

Логи

```
hazelcast2-1 | 2026-02-26 06:59:16,045 [ INFO] [hz.pedantic_bose.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast2]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=7, /172.18.0.6:5701->/172.18.0.8:40190, qualifier=null, endpoint=[172.18.0.8]:40190, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, client name: hz.client_0, client version: 5.6.0
hazelcast1-1 | 2026-02-26 06:59:16,048 [ INFO] [hz.infallible_bhabha.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=7, /172.18.0.4:5701->/172.18.0.8:59610, qualifier=null, endpoint=[172.18.0.8]:59610, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, client name: hz.client_0, client version: 5.6.0
hazelcast3-1 | 2026-02-26 06:59:16,049 [ INFO] [hz.naughty_wozniak.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast3]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=7, /172.18.0.2:5701->/172.18.0.8:56280, qualifier=null, endpoint=[172.18.0.8]:56280, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, client name: hz.client_0, client version: 5.6.0
hazelcast1-1 | 2026-02-26 06:59:16,054 [ INFO] [hz.infallible_bhabha.priority-generic-operation.thread-0] [c.h.i.p.i.PartitionStateManager]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Initializing cluster partition table arrangement...
hazelcast2-1 | 2026-02-26 06:59:50,984 [ INFO] [hz.pedantic_bose.IO.thread-in-0] [c.h.i.s.t.TcpServerConnection]: [hazelcast2]:5701 [dev-cluster] [5.3.8] Connection[id=7, /172.18.0.6:5701->hws-facade-1.hws_default/172.18.0.8:40190, qualifier=null, endpoint=[172.18.0.8]:40190, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hazelcast3-1 | 2026-02-26 06:59:50,984 [ INFO] [hz.naughty_wozniak.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hazelcast3]:5701 [dev-cluster] [5.3.8] Connection[id=7, /172.18.0.2:5701->hws-facade-1.hws_default/172.18.0.8:56280, qualifier=null, endpoint=[172.18.0.8]:56280, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hazelcast1-1 | 2026-02-26 06:59:50,984 [ INFO] [hz.infallible_bhabha.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Connection[id=7, /172.18.0.4:5701->hws-facade-1.hws_default/172.18.0.8:59610, qualifier=null, endpoint=[172.18.0.8]:59610, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
```

```
hazelcast2-1 | 2026-02-26 06:59:50,985 [ INFO] [hz.pedantic_bose.event-4] [c.h.c.i.ClientEndpointManager]: [hazelcast2]:5701 [dev-cluster] [5.3.8] Destroying ClientEndpoint{connection=Connection[id=7, /172.18.0.6:5701->hws-facade-1.hws_default/172.18.0.8:40190, qualifier=null, endpoint=[172.18.0.8]:40190, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=false, connectionType=PYH, planeIndex=-1], clientUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772089156045, latest clientAttributes=null, labels=[]}
hazelcast3-1 | 2026-02-26 06:59:50,985 [ INFO] [hz.naughty_wozniak.event-4] [c.h.c.i.ClientEndpointManager]: [hazelcast3]:5701 [dev-cluster] [5.3.8] Destroying ClientEndpoint{connection=Connection[id=7, /172.18.0.2:5701->hws-facade-1.hws_default/172.18.0.8:56280, qualifier=null, endpoint=[172.18.0.8]:56280, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=false, connectionType=PYH, planeIndex=-1], clientUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772089156049, latest clientAttributes=null, labels=[]}
hazelcast1-1 | 2026-02-26 06:59:50,986 [ INFO] [hz.infallible_bhabha.event-4] [c.h.c.i.ClientEndpointManager]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Destroying ClientEndpoint{connection=Connection[id=7, /172.18.0.4:5701->hws-facade-1.hws_default/172.18.0.8:59610, qualifier=null, endpoint=[172.18.0.8]:59610, remoteUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, alive=false, connectionType=PYH, planeIndex=-1], clientUid=1ee22519-edff-4ae5-9ee9-05ea49f7fa8a, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772089156048, latest clientAttributes=null, labels=[]}
hazelcast2-1 | 2026-02-26 07:00:00,365 [ WARN] [hz.pedantic_bose.cached.thread-2] [c.h.c.i.ClientEngine]: [hazelcast2]:5701 [dev-cluster] [5.3.8] No connection left to client cluster wide 1ee22519-edff-4ae5-9ee9-05ea49f7fa8a for 10001 millis, cleaning resources of the client
hazelcast2-1 | 2026-02-26 07:00:00,366 [ INFO] [hz.pedantic_bose.cached.thread-2] [c.h.t.TransactionManagerService]: [hazelcast2]:5701 [dev-cluster] [5.3.8] Committing/rolling-back live transactions of client, UUID: 1ee22519-edff-4ae5-9ee9-05ea49f7fa8a
hazelcast1-1 | 2026-02-26 07:00:00,366 [ WARN] [hz.infallible_bhabha.cached.thread-2] [c.h.c.i.ClientEngine]: [hazelcast1]:5701 [dev-cluster] [5.3.8] No connection left to client cluster wide 1ee22519-edff-4ae5-9ee9-05ea49f7fa8a for 10001 millis, cleaning resources of the client
hazelcast1-1 | 2026-02-26 07:00:00,367 [ INFO] [hz.infallible_bhabha.cached.thread-2] [c.h.t.TransactionManagerService]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Committing/rolling-back live transactions of client, UUID: 1ee22519-edff-4ae5-9ee9-05ea49f7fa8a
```

Видно запити на з'єднання до кожного з нод.

Результати

```
Starting NO LOCKS test
Final value: 14543
Time: 4.88s

Starting PESSIMISTIC LOCKING test
Final value: 30000
Time: 17.53s

Starting OPTIMISTIC LOCKING test
Final value: 30000
Time: 12.45s
```

Як можна зауважити, імплементація без блокувань не буде давати правильного результату (30к), оскільки гонитва даних (race condition). З іншої сторони, якщо додати синхронізаційний примітив чи механізм, то кінцеве значення буде валідним, але і відповідно збільшився час на виконання.

Крок 7

Ми спостерігаємо втрату даних, якщо не захистити спільну змінну блокуваннями. У нас може бути ситуація, коли три клієнта одночасно зчитують значення 67, і кожен локально додає до неї одиницю і записує 68. Замість того щоб отримати 70 після трьох операцій, ми отримуємо 68, бо дві операції просто перезаписували змінну.

З блокуваннями бачимо, що результати були однакові.

Песимістичне блокування полягає у тому, що клієнт повністю блокує ключ у базі. Інші клієнти будуть чекати у черзі, поки перший не знімить блокування. Це гарантує консистентність даних що бачимо на зображенні вище.

В ту саму чергу, оптимістичне блокування є менш строгим. Тут клієнт не захоплює повністю права на володіння змінною, а працює більш кооперативно: клієнт читає значення, локально міняє, а потім перевіряє чи значення на базі вже хтось встиг оновити. Якщо значення змінилось, то база відхиляє запит і клієнт знову пробує оновити його. В кінці кінців, всі операції будуть збережені і результат буде правильним.

Ми можемо побачити що оптимістичне блокування працює швидше, оскільки три потоки працюють кооперативно (всі не чекають без діла). В той самий час

песимістичне блокування працює приблизно в 1.5 повільніше, бо витрачається час на зняття блокування та захоплення.

Крок 8

Для забезпечення розміру черги я додав наступну конфігурацію до `hazelcast.yaml`:

```
queue:
  bounded_queue:
    max-size: 10
```

Імплементація скрипту знаходиться у файлі `step8_queue.py`.

```
docker compose exec facade uv run python step8_queue.py
```

Логи

З читачами:

```
troubles=null, labels=[]}
hazelcast3-1 | 2026-02-26 17:08:56,035 [ INFO] [hz.blissful_swanson.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast3]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=18, /172.18.0.3:5701->/172.18.0.8:56574, qualifier=null, endpoint=[172.18.0.8]:56574, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 15a44908-7204-4494-b591-88e35e1d0f3f, client name: hz.client_0, client version: 5.6.0
hazelcast1-1 | 2026-02-26 17:08:56,037 [ INFO] [hz.relaxed_jang.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=18, /172.18.0.4:5701->/172.18.0.8:50944, qualifier=null, endpoint=[172.18.0.8]:50944, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 15a44908-7204-4494-b591-88e35e1d0f3f, client name: hz.client_0, client version: 5.6.0
hazelcast2-1 | 2026-02-26 17:08:56,038 [ INFO] [hz.vigorous_fermat.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast2]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=18, /172.18.0.6:5701->/172.18.0.8:41410, qualifier=null, endpoint=[172.18.0.8]:41410, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 15a44908-7204-4494-b591-88e35e1d0f3f, client name: hz.client_0, client version: 5.6.0
hazelcast3-1 | 2026-02-26 17:08:56,085 [ INFO] [hz.blissful_swanson.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hazelcast3]:5701 [dev-cluster] [5.3.8] Connection[id=18, /172.18.0.3:5701->/172.18.0.8:56574, qualifier=null, endpoint=[172.18.0.8]:56574, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hazelcast1-1 | 2026-02-26 17:08:56,085 [ INFO] [hz.relaxed_jang.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Connection[id=18, /172.18.0.4:5701->/172.18.0.8:50944, qualifier=null, endpoint=[172.18.0.8]:50944, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hazelcast3-1 | 2026-02-26 17:08:56,085 [ INFO] [hz.blissful_swanson.event-5] [c.h.c.i.ClientEndpointManager]: [hazelcast3]:5701 [dev-cluster] [5.3.8] Destroying ClientEndpoint{connection=Connection[id=18, /172.18.0.3:5701->/172.18.0.8:56574, qualifier=null, endpoint=[172.18.0.8]:56574, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=false, connectionType=PYH, planeIndex=-1], clientUid=15a44908-7204-4494-b591-88e35e1d0f3f, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772125736035, latestClientAttributes=null, labels=[]}
hazelcast2-1 | 2026-02-26 17:08:56,085 [ INFO] [hz.vigorous_fermat.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hazelcast2]:5701 [dev-cluster] [5.3.8] Connection[id=18, /172.18.0.6:5701->/172.18.0.8:41410, qualifier=null, endpoint=[172.18.0.8]:41410, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hazelcast1-1 | 2026-02-26 17:08:56,085 [ INFO] [hz.relaxed_jang.event-5] [c.h.c.i.ClientEndpointManager]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Destroying ClientEndpoint{connection=Connection[id=18, /172.18.0.4:5701->/172.18.0.8:50944, qualifier=null, endpoint=[172.18.0.8]:50944, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=false, connectionType=PYH, planeIndex=-1], clientUid=15a44908-7204-4494-b591-88e35e1d0f3f, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772125736036, latestClientAttributes=null, labels=[]}
hazelcast2-1 | 2026-02-26 17:08:56,085 [ INFO] [hz.vigorous_fermat.event-4] [c.h.c.i.ClientEndpointManager]: [hazelcast2]:5701 [dev-cluster] [5.3.8] Destroying ClientEndpoint{connection=Connection[id=18, /172.18.0.6:5701->/172.18.0.8:41410, qualifier=null, endpoint=[172.18.0.8]:41410, remoteUid=15a44908-7204-4494-b591-88e35e1d0f3f, alive=false, connectionType=PYH, planeIndex=-1], clientUid=15a44908-7204-4494-b591-88e35e1d0f3f, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772125736038, latestClientAttributes=null, labels=[]}
```

Без читачів:

```
hazelcast1-1 | 2026-02-26 17:24:52,283 [ INFO] [hz.relaxed_jang.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast1]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=20, /172.18.0.4:5701->/172.18.0.8:44236, qualifier=null, endpoint=[172.18.0.8]:44236, remoteUid=3a635964-0dcb-4619-9334-57eab5f63352, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 3a635964-0dcb-4619-9334-57eab5f63352, client name: hz.client_0, client version: 5.6.0
hazelcast3-1 | 2026-02-26 17:24:52,284 [ INFO] [hz.blissful_swanson.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast3]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=20, /172.18.0.3:5701->/172.18.0.8:39260, qualifier=null, endpoint=[172.18.0.8]:39260, remoteUid=3a635964-0dcb-4619-9334-57eab5f63352, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 3a635964-0dcb-4619-9334-57eab5f63352, client name: hz.client_0, client version: 5.6.0
hazelcast2-1 | 2026-02-26 17:24:52,285 [ INFO] [hz.vigorous_fermat.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hazelcast2]:5701 [dev-cluster] [5.3.8] Received auth from Connection[id=20, /172.18.0.6:5701->/172.18.0.8:39062, qualifier=null, endpoint=[172.18.0.8]:39062, remoteUid=3a635964-0dcb-4619-9334-57eab5f63352, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 3a635964-0dcb-4619-9334-57eab5f63352, client name: hz.client_0, client version: 5.6.0
```

Бачимо, що присутній запит від клієнта: Received auth from Connection.

Результати

<pre>[Producer] Starting... [Reader 1] Ready for reading [Reader 2] Ready for reading [Producer] Wrote: 1 -> [Reader 1] Read: 1 -> [Reader 2] Read: 2 [Producer] Wrote: 2 -> [Reader 1] Read: 3 [Producer] Wrote: 3 -> [Reader 2] Read: 4 [Producer] Wrote: 4 -> [Reader 1] Read: 5 [Producer] Wrote: 5 -> [Reader 2] Read: 6 [Producer] Wrote: 6 -> [Reader 1] Read: 7 [Producer] Wrote: 7 -> [Reader 2] Read: 8 [Producer] Wrote: 8 -> [Reader 1] Read: 9 [Producer] Wrote: 9 [Producer] Wrote: 10 -> [Reader 2] Read: 10 -> [Reader 1] Read: 11 [Producer] Wrote: 11 -> [Reader 2] Read: 12 [Producer] Wrote: 12 [Producer] Wrote: 13 -> [Reader 1] Read: 13 -> [Reader 2] Read: 14 [Producer] Wrote: 14 -> [Reader 1] Read: 15 [Producer] Wrote: 15 [Producer] Wrote: 16 -> [Reader 2] Read: 16 -> [Reader 1] Read: 17 [Producer] Wrote: 17 -> [Reader 2] Read: 18 [Producer] Wrote: 18 [Producer] Wrote: 19 -> [Reader 1] Read: 19 -> [Reader 2] Read: 20 [Producer] Wrote: 20 -> [Reader 1] Read: 21 [Producer] Wrote: 21 -> [Reader 2] Read: 22 [Producer] Wrote: 22 -> [Reader 1] Read: 23 [Producer] Wrote: 23 [Producer] Wrote: 24 -> [Reader 2] Read: 24 -> [Reader 1] Read: 25 [Producer] Wrote: 25 [Producer] Wrote: 26 -> [Reader 2] Read: 26 [Producer] Wrote: 26 -> [Reader 1] Read: 27 [Producer] Wrote: 27 -> [Reader 2] Read: 28 [Producer] Wrote: 28</pre>	<pre>-> [Reader 1] Read: 29 [Producer] Wrote: 29 -> [Reader 2] Read: 30 [Producer] Wrote: 31 -> [Reader 1] Read: 31 [Producer] Wrote: 32 -> [Reader 2] Read: 32 [Producer] Wrote: 33 -> [Reader 1] Read: 33 [Producer] Wrote: 34 -> [Reader 2] Read: 34 [Producer] Wrote: 35 -> [Reader 1] Read: 35 [Producer] Wrote: 36 -> [Reader 2] Read: 36 [Producer] Wrote: 37 -> [Reader 1] Read: 37 [Producer] Wrote: 38 -> [Reader 2] Read: 38 [Producer] Wrote: 39 -> [Reader 1] Read: 39 [Producer] Wrote: 40 -> [Reader 2] Read: 40 [Producer] Wrote: 41 -> [Reader 1] Read: 41 [Producer] Wrote: 42 -> [Reader 2] Read: 42 [Producer] Wrote: 43 -> [Reader 1] Read: 43 -> [Reader 2] Read: 44 [Producer] Wrote: 44 -> [Reader 1] Read: 45 [Producer] Wrote: 45 [Producer] Wrote: 46 -> [Reader 2] Read: 46 [Producer] Wrote: 47 -> [Reader 1] Read: 47 [Producer] Wrote: 48 -> [Reader 2] Read: 48 [Producer] Wrote: 49 -> [Reader 1] Read: 49 [Producer] Wrote: 50 [Producer] Wrote: 51 -> [Reader 2] Read: 51 [Producer] Wrote: 52 -> [Reader 1] Read: 52 [Producer] Wrote: 53 -> [Reader 2] Read: 53 [Producer] Wrote: 54 [Producer] Wrote: 55 -> [Reader 1] Read: 55 [Producer] Wrote: 56 -> [Reader 2] Read: 56 [Producer] Wrote: 57 -> [Reader 1] Read: 57 [Producer] Wrote: 58 -> [Reader 2] Read: 58 [Producer] Wrote: 59 [Producer] Wrote: 60 -> [Reader 2] Read: 60</pre>	<pre>[Producer] Wrote: 61 -> [Reader 1] Read: 61 [Producer] Wrote: 62 -> [Reader 2] Read: 62 [Producer] Wrote: 63 -> [Reader 1] Read: 63 [Producer] Wrote: 64 -> [Reader 2] Read: 64 [Producer] Wrote: 65 -> [Reader 1] Read: 65 [Producer] Wrote: 66 -> [Reader 2] Read: 66 [Producer] Wrote: 67 -> [Reader 1] Read: 67 [Producer] Wrote: 68 -> [Reader 2] Read: 68 [Producer] Wrote: 69 -> [Reader 1] Read: 69 [Producer] Wrote: 70 -> [Reader 2] Read: 70 [Producer] Wrote: 71 -> [Reader 1] Read: 71 [Producer] Wrote: 72 -> [Reader 2] Read: 72 [Producer] Wrote: 73 -> [Reader 1] Read: 73 [Producer] Wrote: 74 -> [Reader 2] Read: 74 [Producer] Wrote: 75 -> [Reader 1] Read: 75 [Producer] Wrote: 76 [Producer] Wrote: 77 -> [Reader 2] Read: 77 [Producer] Wrote: 78 [Producer] Wrote: 79 -> [Reader 1] Read: 79 [Producer] Wrote: 80 -> [Reader 2] Read: 80 [Producer] Wrote: 81 -> [Reader 1] Read: 81 [Producer] Wrote: 82 [Producer] Wrote: 83 -> [Reader 2] Read: 83 [Producer] Wrote: 84 [Producer] Wrote: 85 -> [Reader 1] Read: 85 [Producer] Wrote: 86 [Producer] Wrote: 87 -> [Reader 2] Read: 87 [Producer] Wrote: 88 [Producer] Wrote: 89 -> [Reader 1] Read: 89 [Producer] Wrote: 90 [Producer] Wrote: 91 -> [Reader 2] Read: 91 [Producer] Wrote: 92</pre>	<pre>[Producer] Wrote: 93 -> [Reader 1] Read: 93 -> [Reader 2] Read: 94 [Producer] Wrote: 94 [Producer] Wrote: 95 -> [Reader 1] Read: 95 -> [Reader 2] Read: 96 [Producer] Wrote: 96 -> [Reader 1] Read: 97 [Producer] Wrote: 97 -> [Reader 2] Read: 98 [Producer] Wrote: 98 [Producer] Wrote: 99 -> [Reader 1] Read: 99 [Producer] Wrote: 100 -> [Reader 2] Read: 100</pre>	<pre>[Producer] Starting... [Producer] Wrote: 1 [Producer] Wrote: 2 [Producer] Wrote: 3 [Producer] Wrote: 4 [Producer] Wrote: 5 [Producer] Wrote: 6 [Producer] Wrote: 7 [Producer] Wrote: 8 [Producer] Wrote: 9 [Producer] Wrote: 10</pre>
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На лівій частині бачимо що черга працює коректно та вичитка відбувається одразу. На крайній правій правій фотці бачимо що коли черга переповнюється програма блокується, продюсер чекає поки звільниться у черзі місце.

3 читачами:

Queue Statistics						RESET METRICS	Now
Member ^	Items ^	Backups ^	Max Age ^	Min Age ^	Average Age ^		
hazelcast1:5701	N/A	N/A	N/A	N/A	N/A		
hazelcast2:5701	0	0	1m 28s 238ms	1ms	11s 924ms		
hazelcast3:5701	0	0	1m 28s 238ms	1ms	8s 404ms		

1 - 3 of 3 Rows 10

Без читачів:

Queue Statistics						RESET METRICS	Now
Member ^	Items ^	Backups ^	Max Age ^	Min Age ^	Average Age ^		
hazelcast1:5701	N/A	N/A	N/A	N/A	N/A		
hazelcast2:5701	0	10	1m 28s 238ms	1ms	11s 924ms		
hazelcast3:5701	10	0	1m 28s 238ms	1ms	8s 404ms		

1 - 3 of 3 Rows 10

Кожне повідомлення з черги вичитується тільки одним клієнтом. Як тільки елемент вичитується методом `take()`, то він видаляється з черги та другий клієнт не може його побачити. У логах вище можна побачити що числа не повторюються та розподіляються між клієнтами майже порівну. Бачимо з першого фото менеджмент центру, що в черзі не залишилось жодного елемента.

З іншої сторони, при вилученні читачів з програми, черга перестає наповнюватися після 10-го елемента та скрипт призупиняє роботу. Це стається, оскільки ми встановили обмеження на 10 елементів `max-size: 10` і відповідно поведінка буде блокуюча в разі переповнення. Продюсер бужде нескінченно чекати поки хоча б один читач не вчитає щось. (цього не відбудеться). На зображенні нижче від попереднього бачимо, що розмір черги відповідний.

Також у Hazelcast, елементи черги знаходяться на одній ноді та не розподіляються між ними. Кожний вузол має окрему роль, він або зберігає елементи черги, або бекап. Третій в цьому випадку є лишнім та позначається як N/A.

Висновок

У цьому завданні познайомився з роботою Hazelcast та на практиці спробував Distributed Map та Bounded Queue. Також навчився конфігурувати ноди та робити окремі налаштування для структур даних.

Джерела

- [1]: [Default backup-count](#)